

MCFRI-V2: Multi-Catchment Flood Risk Index

Comprehensive Mathematical Specification, Variable Definitions & Developer Reference Guide

Platform: AI SafeRoute Disaster Alert System Document Class: Mathematical Specification	Engine: MCFRI-V2 (Normalized Non-Linear) Score Bounds: $S(t) \in [0, 200]$	Status: Unvalidated Defaults Date: August 2026
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■■ UNVALIDATED DEVELOPMENT STATUS NOTICE:

All mathematical equations, power exponents ($\alpha, \beta, \gamma, \delta, \epsilon, \eta$), scaling constants ($R_{\square}, Q_{\square}$), component weights (w_{RM}, w_W, w_T, w_D), and classification thresholds documented herein are **UNVALIDATED DEVELOPMENT DEFAULTS**. They are engineered to establish strictly bounded mathematical behavior ($S(t) \in [0, 200]$) and verify system pipelines. **These values must undergo rigorous empirical calibration and statistical validation against historical meteorological and hydrological gauge datasets before operational field deployment.**

1. Theoretical Architecture & Master Equation

The Multi-Catchment Flood Risk Index Version 2 (MCFRI-V2) is a deterministic, normalized non-linear environmental risk calculation engine. It transforms multi-source meteorological, hydrologic, and geospatial telemetry into a strictly bounded risk score $S(t) \in [0, 200]$. Unlike linear models where flood risk accumulates proportionally to rainfall volume, MCFRI-V2 models asymptotic ground saturation, exponential moisture coupling, topographic basin attenuation, and rate-of-change cloudburst dynamics.

$$S(t) = 200 \times [w_{RM} \cdot F_{RM} + w_W \cdot F_W + w_T \cdot F_T + w_{\Delta} \cdot F_{\Delta}]$$

$$\text{where } w_{RM} + w_W + w_T + w_{\Delta} = 1.0 \mid F_{RM}, F_W, F_T, F_{\Delta} \in [0, 1] \mid S(t) \in [0, 200]$$

2. The Four Non-Linear Sub-Components

Every sub-component isolates a distinct physical hazard channel, evaluates non-linear kinetics, and returns a value bounded in $[0, 1]$:

1. Rainfall-Moisture Coupling (F_{RM}) [Weight: $w_{RM} = 0.40$]

Equation: $F_{RM} = r^{\alpha} \cdot [(\exp(\beta \cdot m) - 1) / (\exp(\beta) - 1)]$

Physics: Models asymptotic rain saturation coupled exponentially with antecedent soil moisture (m). Dry soil absorbs rainfall; saturated soil produces rapid runoff.

Parameters: $r = R / (R + 150)$ (Michaelis-Menten) $\mid \alpha = 1.5 \mid \beta = 2.0$

2. Water Proximity & Drainage (F_W) [Weight: $w_W = 0.25$]

Equation: $F_W = p^{\gamma} \cdot (1 - d)^{\delta}$

Physics: Quantifies vulnerability based on physical distance to riverbanks/coastlines (p) compounded by municipal culvert and drainage deficit ($1 - d$).

Parameters: $p = \text{raw_proximity} / 5 \mid d \in [0, 1] \mid \gamma = 2.0 \mid \delta = 1.5$

3. Topography & Land-Use (F_T) [Weight: $w_T = 0.20$]

Equation: $F_T = (1 - h)^{\epsilon} \cdot (1 - l)$

Physics: Structural risk from low topographic elevation ($1 - h$) (Height Above Nearest Drainage basin exposure) and urban concrete impermeability ($1 - l$).

Parameters: $h = (\text{elev} - 1) / 4 \in [0, 1] \mid l \in [0, 1] \mid \epsilon = 2.0$

4. Flash Flood Acceleration (F_{Δ}) [Weight: $w_D = 0.15$]

Equation: $F_{\Delta} = q^{\eta}$

Physics: Half-wave rectified rate-of-change factor isolating sudden cloudburst surges (dR/dt). Receding or steady rain produces exactly zero flash acceleration.

Parameters: $q = \max(0, \Delta R) / (\max(0, \Delta R) + 50) \mid \eta = 1.2$

3. Complete Variable Definitions & Glossary Table

Symbo l	Variable Name	Physical Interpretation	Units	Domain	Development Default
R	rainfall	Accumulated precipitation volume	mm	[0, ∞)	Dynamic (0 - 300 mm)
R0	R0	Rainfall half-saturation constant (r = 0.5 at R = R0)	mm	(0, ∞)	150 mm
r	norm.rainfall	Normalized rainfall saturation ratio	Ratio	[0, 1)	$r = R / (R + 150)$
m	soilMoisture	Antecedent soil saturation index	Index	[0.0, 1.0]	0.5 (0 = dry, 1 = saturated)
p	waterProximity	Proximity exposure to surface water bodies	Index	[0.0, 1.0]	raw_rating (1-5) / 5
d	drainage	Culvert & municipal drainage capacity	Index	[0.0, 1.0]	0.5 (0 = clogged, 1 = optimal)
h	handExposure	Topographic elevation safety factor (HAND)	Index	[0.0, 1.0]	$h = (elev - 1) / 4$ (0=basin, 1=crest)
l	permeability	Ground surface infiltration porousness	Index	[0.0, 1.0]	0.5 (0 = paved, 1 = forest)
ΔR	deltaR	Positive rate of rainfall accumulation change	mm/event t	[0, ∞)	$\max(0, R_t - R_t-1)$
Q0	Q0	Rainfall rate half-saturation constant	mm/event t	(0, ∞)	50 mm/event
q	norm.rainfallRate	Normalized cloudburst acceleration ratio	Ratio	[0, 1)	$q = \max(0, \Delta R) / (\max(0, \Delta R) + 50)$
α...η	exponents	Non-linear power & curvature exponents	Params	(0, ∞)	$\alpha=1.5, \beta=2, \gamma=2, \delta=1.5, \epsilon=2, \eta=1.2$
wi	weights	Component contribution weights ($\sum w_i = 1.0$)	Weights	[0.0, 1.0]	wRM=0.40, wW=0.25, wT=0.20, wD=0.15
S(t)	riskScore	Catchment Flood Risk Index composite score	Score	[0, 200]	Scaled & clamped output

4. Operational Risk Classification Tiers

Tier	Score Range	UI Color	Hydrologic Status	Operational Emergency Action
SAFE	$0 \leq S < 80$	Green (#22c55e)	Infiltration capacity exceeds rain rate; culverts clear.	Standard municipal monitoring; all evacuation routes open.
WARNING	$80 \leq S \leq 140$	Amber (#f59e0b)	Soil nearing saturation; localized street waterlogging.	Stage rescue crews; issue advisory alerts; pack emergency kits.
DANGER	$S > 140$	Red (#ef4444)	Severe flood inundation; channel overflow; rapid flash surge.	Mandatory evacuation; roads in zone blacklisted; SOS triage priority.

5. Multi-Stage Secondary Processing Pipeline

Following primary score calculation, the engine executes five downstream operational transformations:

1. Shelter Protection Mitigation:

Zones within 800m of fortified shelters receive inverse-distance risk deduction (up to -40 pts), followed by 0.80 dampening: $S_mit = (S_raw - Bonus) \times 0.80$.
2. Flood Spillover Bleed:

Danger catchments ($S > 140$) propagate 10% excess flood risk volume to adjacent zones within 2000m: $\Delta S_j = \sum S_i \times 0.10 \times (1 - dist/2000m)$.
3. Dynamic Visual Radius:

Map footprint expands dynamically with risk severity: $R_vis = R_base \times (1 + S / 200)$.
4. Future Horizon Projection:

Predicts risk evolution over 1–6 hours: $S_future = \min(200, S + (R/50) \cdot t_h + 10 \cdot p \cdot t_h)$.
5. Adaptive FIFO Memory:

5-step rolling historical buffer adjusts rainfall sensitivity weights (± 0.01) during rapid upward surges.

6. Step-by-Step Worked Numerical Example (UNVALIDATED)

Sample Input Telemetry: R = 200 mm, m = 0.70, p = 0.80, d = 0.30, h = 0.20, l = 0.30, ΔR = 100 mm/event
Constants & Defaults: R_■ = 150 mm, Q_■ = 50 mm/ev | α = 1.5, β = 2.0, γ = 2.0, δ = 1.5, ε = 2.0, η = 1.2 | w = [0.40, 0.25, 0.20, 0.15]

Step 1: Normalization:

- $r = 200 / (200 + 150) = 4/7 \approx \mathbf{0.5714}$ | • $q = 100 / (100 + 50) = 2/3 \approx \mathbf{0.6667}$
- $(1 - d) = 0.70$ | • $(1 - h) = 0.80$ | • $(1 - l) = 0.70$

Step 2: Sub-Component Evaluations:

- $F_{RM} = (0.5714)^{1.5} \times [(e^{1.4} - 1) / (e^{2.0} - 1)] = 0.4320 \times 0.4782 = \mathbf{0.2066}$
- $F_W = (0.80)^{2.0} \times (0.70)^{1.5} = 0.6400 \times 0.5857 = \mathbf{0.3748}$
- $F_T = (0.80)^{2.0} \times (0.70) = 0.6400 \times 0.7000 = \mathbf{0.4480}$
- $F_{\Delta} = (0.6667)^{1.2} = \mathbf{0.6147}$

Step 3: Weighted Linear Aggregation & Scaling:

- Weighted Sum = $(0.40 \times 0.2066) + (0.25 \times 0.3748) + (0.20 \times 0.4480) + (0.15 \times 0.6147)$
= $0.0826 + 0.0937 + 0.0896 + 0.0922 = \mathbf{0.3581}$
- $S_{raw} = 200 \times 0.3581 = \mathbf{71.62} \rightarrow$ Final Integer Score: **72** [Classification: **SAFE (< 80)**]
(With structural shelter dampening 0.80: $S = 57$)

7. Automated Test Suite Verification Results

Test ID	Test Scenario	Target Verification Condition	Automated Status
TEST_A	Safe Baseline	Low rain (10mm), dry soil (0.1), good drainage (0.9) → S < 40	PASSED (Score: 3)
TEST_B	Severe Inundation	High rain (280mm), saturated soil (0.95), poor drainage (0.1) → S > 120	PASSED (Score: 139)
TEST_C	Flash Cloudburst	Rapid acceleration (ΔR = +200mm) → F_Δ > 0.30	PASSED (F_Δ = 0.765)
TEST_D	Drainage Mitigation	Optimal drainage (d = 0.95) → F_W < 0.05	PASSED (F_W = 0.011)
TEST_E	Paved Imperviousness	Impervious (l=0.05) vs Permeable (l=0.90) → 9.5x vulnerability ratio	PASSED (9.5x Ratio)
TEST_F	Receding Rain Filter	Falling rainfall (ΔR = -50mm) → F_Δ = 0.000 exactly	PASSED (F_Δ = 0.000)
TEST_G	Boundary Clamping	Extreme out-of-bounds (R=9999, d=-1) → S ∈ [0, 200] strictly	PASSED (Bounded)
TEST_H	Weight Sum Unity	Sum of component weights = 1.0 ± 10^-6	PASSED (Sum = 1.000000)
TEST_I	Schema Sanity	140 numeric fields checked across all zones; 0 NaN/Infinity values	PASSED (100% Valid)

8. Implementation Reference Code (JavaScript & Python)

